

Voltage Sag Estimation

Leonardo Gabriel Granda Erazo

Electrical Engineering Program

Universidad Nacional de Loja

Faculty of Energy, Industry and Non-Renewable Natural Resources

leonardo.granda@unl.edu.ec

Abstract—This report addresses the simulation and assessment of the voltage sags that can occur at a substation, their consequences, and how they affect power quality (PQ). The simulation was carried out using Matlab and Simulink.

Index Terms—Voltage sags, substation, power quality, Simulink.

I. INTRODUCTION

Voltage sags in distribution systems are faults that can affect the generating and distribution utilities and the end users; their main effect is economic, since the process of resolving them is very costly. It is often assumed that the main cause of a voltage sag at a substation lies in the transmission system, but, as shown below, the main faults leading to these sags are those occurring on the parallel feeders.

II. PROBLEM STATEMENT

Fig. 1 shows the outgoing busbar of a substation at 22 kV line-to-line, 50 Hz, with three distribution feeders. The three-phase short-circuit level at the point of common coupling (PCC) is 100 MVA with an X/R ratio of 7. The per-kilometer line impedance can be considered identical on all feeders, with an X/R ratio equal to that of the source impedance. In addition, the total impedance of the longest feeder equals 1.08 times the source impedance.

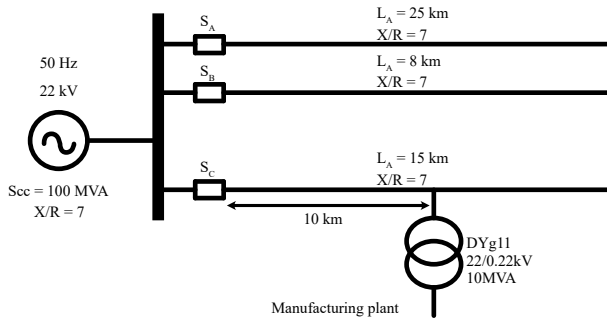


Fig. 1. Graphical representation of the problem.

The substation breakers S_A , S_B , and S_C clear single-line-to-ground (SLG) faults on feeders A, B, and C in 6, 2.5, and 6 cycles, respectively. The fault-performance data show that feeder B (mainly an overhead line) has an SLG fault rate of 0.3 events/km per year, while feeders A and C have an SLG fault rate of 0.1 events/km per year.

SLG faults in the distribution system have been blamed for most of the PQ problems at an assembly plant fed from

feeder C and located 10 km from the substation. This plant is supplied through a DYg11 22/0.22 kV transformer, and severe problems are occurring due to the tripping of several single-phase drives, similar to those studied in the previous assignment (here the system is sensitive to sags below 80 V_{RMS}), connected randomly across the phases (phase-to-neutral). Each event that causes the malfunction of one or more of these devices produces an estimated financial loss of \$6k due to lost materials, production, and wages.

The task is to estimate the total annual financial losses caused by PQ problems at the plant, through the following steps:

- 1) Determine the vulnerability area of the plant (trial and error).
- 2) Estimate the expected number of faults within the vulnerability area.
- 3) Estimate the total annual financial losses from the given data.
- 4) Verify the validity of the results.

III. BACKGROUND

Simulating this workshop in Simulink required values not provided by the problem, so Matlab was used [1]. One of the first steps was determining the feeder impedances, using (1), (2), and (3).

$$S = \frac{V^2}{z} \quad (1)$$

$$|z| = \sqrt{R^2 + X^2} \quad (2)$$

$$\frac{X}{R} = 7 \quad (3)$$

Solving these expressions gives the source impedance $z_{source} = (0.6845 + 4.7914j) \Omega$. Since the longest feeder has 1.08 times the source impedance, the impedance of the longest feeder—feeder A, at 25 km—is $z_{SA} = (0.7392 + 5.1747j) \Omega$; as this is a function of distance, the per-kilometer impedance is $z_{\Omega/km} = (0.0296 + 0.207j) \Omega/km$. From this expression, the impedances of the other feeders and their resistances and inductances are obtained.

To analyze the faults, a fault distance is needed, determined here by trial and error. With the fault distance, the fault impedance is obtained, which splits the feeder into a pre-fault and a post-fault section—two distinct impedances that

depend on the remaining distance. This also allows the load's vulnerability area to be determined, i.e., the distances over which the load can keep operating regardless of being at the minimum voltage. Once the impedances are known, the corresponding blocks are connected in Simulink.

To estimate the number of faults within the vulnerability area, assuming the fault occurs in the distribution system, (4) is applied, where $E_{Parallel}$ is the estimated number of faults on the parallel feeders, N is the SLG fault rate, and E_P is the feeder length within the vulnerability area (subscripts 1 and 3 denote feeders A and C).

$$E_{Parallel} = N_1 E_{P1} + N_3 E_{P3} \quad (4)$$

IV. SIMULATION DEVELOPMENT

A. Fault on Feeder A

Fig. 2 shows that, when the fault occurs (here at 0.08 s), the first voltage affected is the transformer voltage; consequently, the load voltage begins to decay. After some time—specifically at 0.12 s—the transformer voltage returns to normal and the load voltage stabilizes.

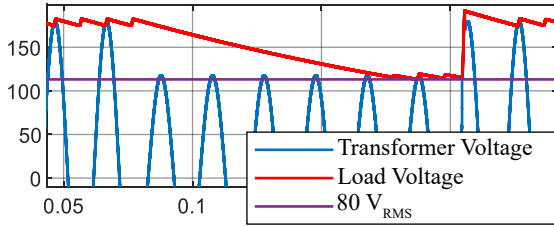


Fig. 2. Fault on feeder A.

As in the previous assignment, when a voltage sag occurs the DC bus begins to deliver part of its stored energy to the load, but this does not prevent the sag because of the capacitor discharge rate; this is why the load-voltage curve falls before eventually recovering. Here the load voltage does not drop below 80 V_{RMS}, i.e., it sits at the edge of the vulnerability area. From the trial-and-error process, a fault was found to be critical if it occurs within the first 6.6329 km of feeder A. Fig. 3 shows the load voltage at this limiting distance.

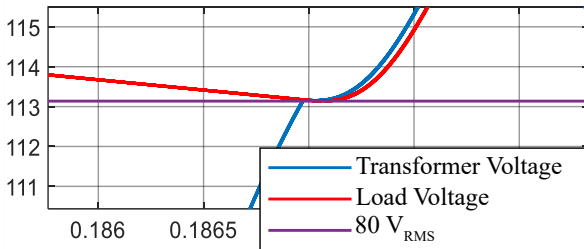


Fig. 3. Fault on feeder A at 6.6329 km.

B. Fault on Feeder B

Fig. 4 shows a voltage sag at the load and transformer, but unlike the previous case, over the entire 8 km of the feeder—and even beyond—the effect on the load is not as harmful: even with a fault on this feeder, the load can keep operating without crossing the limit voltage.

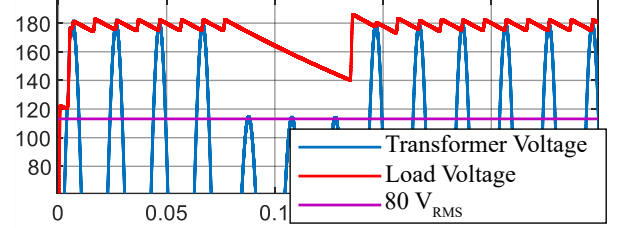


Fig. 4. Fault on feeder B.

This is partly due to the recovery time of feeder B, the fastest of the three to clear. Because of this response time, the load does not fall below 80 V_{RMS}: the load could keep decaying, but since the feeder recovers in 0.05 s, the load and transformer stabilize and return to normal. Testing several distances confirmed that feeder B is the least harmful, so there is no vulnerability area for the load on feeder B, since it never experiences a sag considered dangerous.

C. Fault on Feeder C

Fig. 5 shows what would happen with a fault on feeder C. Unlike the previous case, here the load does exceed the limit value. Over the trial-and-error range of 0 to 15 km, the load voltage crossed the limit in every test. Although a faster reconnection would avoid serious consequences, this feeder takes 0.12 s, giving the load more time to keep sagging, so by the time the feeder recovers the load has already been disturbed. The load is therefore at risk over the entire length of feeder C: any fault on this feeder produces a sag that exceeds 80 V_{RMS}.

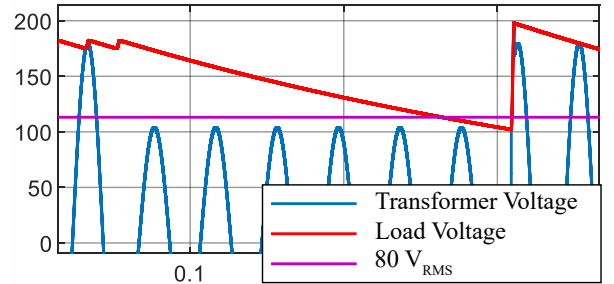


Fig. 5. Fault on feeder C.

V. RESULTS

A. Vulnerability Area

For feeder A, the distance over which the load can keep operating at the limit is approximately 6.6329 km; since it

was determined by trial and error, it cannot be guaranteed to be exact. For feeder B, a fault causes no adverse effects on the load, so the load is not vulnerable to significant sags there. For feeder C, a fault at any point along its coverage causes the load to drop below 80 V_{RMS}, so the load is vulnerable over the full 15 km. Fig. 6 shows an approximate diagram of the vulnerability area.

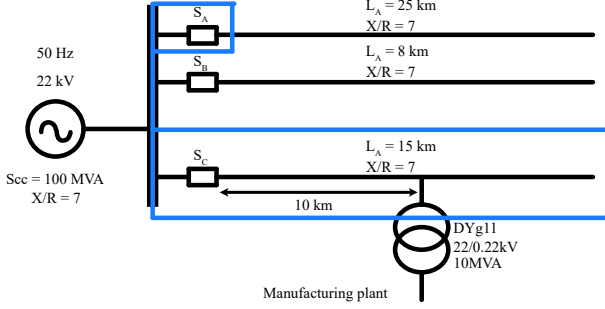


Fig. 6. Vulnerability area of the load.

B. Estimation of the Number of Faults Within the Vulnerability Area

Applying (4) with the fault rates and vulnerable lengths of the three feeders (feeder B contributes zero, as it has no vulnerability area):

$$E_{\text{Parallels}} = 0.1(6.6329) + 0.3(0) + 0.1(15) = 2.16329 \text{ faults}$$

Thus, the estimated number of faults within the vulnerability area per year is 2.16329.

C. Total Annual Financial Losses

Since each event causes a \$6k loss, the financial losses are:

$$P_{\text{fin}} = 6000 \frac{\$}{\text{fault}} \times 2.16329 \text{ faults} = \$12,979.74$$

The plant can therefore expect financial losses of about \$13k per year.

D. Validation of the Results

The only feeders with a clearly appreciable vulnerability area are A and C, so the following figures validate the results for these two.

1) *Feeder A performance:* To confirm proper operation outside the vulnerability area (which reaches 6.6329 km on this feeder), the simulation was run for a fault at 15 km, giving Fig. 7. Outside the vulnerability area, the load voltage does not cross the limit; the farther the fault from 6.6329 km, the safer the load. Conversely, if the fault occurs before 6.6329 km, the load crosses the protection limit, as shown in Fig. 8.

2) *Feeder C performance:* For feeder C, the simulation was also run beyond the vulnerability area (its full 15 km), i.e., at a fault distance of 30 km, giving Fig. 9. This shows that, regardless of the feeder length or the fault location, feeder C—being the unit that supplies the load—always leaves the load at risk of falling below 80 V_{RMS}.

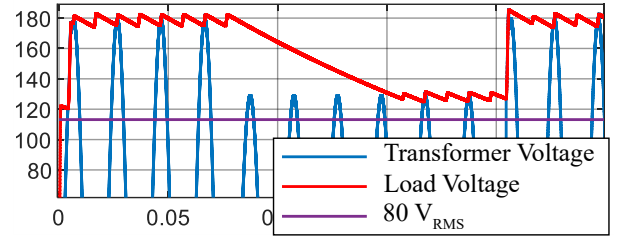


Fig. 7. Fault on feeder A at 15 km.

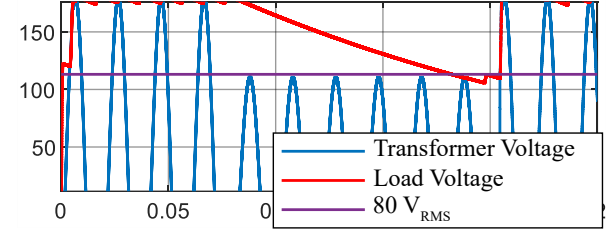


Fig. 8. Fault on feeder A at 3 km.

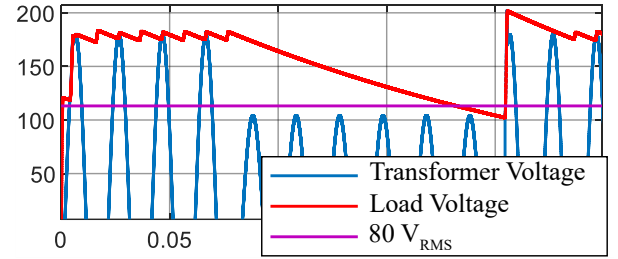


Fig. 9. Fault on feeder C at 30 km.

VI. CONCLUSIONS

The analysis and simulation showed what can happen to a load supplied by a feeder that runs in parallel with others when one of those feeders faults. Because they are in parallel, a fault on one—such as a voltage sag or an interruption—can affect the rest, with consequences that depend on the protection devices and on how stable the feeders are.

Trial and error confirmed that, on feeder A, the load is susceptible to falling below the limit when the fault occurs within the first 6.6329 km; beyond that distance, a fault does not cause a significant sag. Feeder C, being the unit the load is connected to, affects the load's stability for a fault at any distance along its coverage. This does not happen on feeder B: a fault there does cause a sag, but not deep enough to cross the limit, so the load keeps operating normally.

A DC bus can supply the load for a limited time during a sag, but this is not decisive here, because the feeders have reconnection protocols that guarantee the sag is cleared within a set time. This clearing time is what distinguishes the feeders: feeders A and C recover after 6 cycles (0.12 s), while feeder B recovers in 2.5 cycles (0.05 s). Feeder B therefore causes no major adverse effects on the load thanks to its fast recovery;

feeder A has a limiting distance within which the load remains at risk, and feeder C puts the load at serious risk for any fault.

A second conclusion follows: in the distribution system, the voltage sag depends on how long a feeder takes to clear the fault. This is reflected in the ITI and CBEMA curves, which specify the sag voltage at which certain equipment can keep operating as a function of the fault duration in cycles—the more cycles the fault lasts, the deeper the sag. In fact, the sag on feeder A would be the same as on feeder B if the latter took longer to clear.

Finally, these faults have economic effects on companies, which underscores the importance of good PQ. Faults can arise from causes beyond the reach of users or the responsible entities, and reconnection protocols exist to clear them as quickly as possible and ensure good service; otherwise, the main negative effect is economic. In this workshop, with a rate of about 2.16 faults per year, the plant would have to pay nearly \$13k.

REFERENCES

- [1] MathWorks. [Online]. Available: <https://matlab.mathworks.com/>